

BG04

Fibre Optic Based Vibration Sensing: Nature of the Measurement

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SUMMARY

Within the industry there appears to be some confusion about what fibre optic based seismic systems are actually measuring. Using synthetic and real data we show that such systems measure instantaneous strain, which is equivalent to the sum of the perturbations of each section of the gauge length from its mean position divided by its length. The time derivative of strain, the strain rate, is proportional to the average velocity measured at all points along the gauge length. DVS measurements have the polarity of a hydrophone measurement but the directionality is highly complex and depends on wavenumber as well as the angle of incidence.

Introduction

Interest in the use of fibre-optic based distributed vibration sensing (DVS) systems is rapidly expanding with users primarily drawn by its ability to make a large number of high-density measurements without the need for individual sensors. In DVS surveys the fibre-optic cable itself is used as a sensor; pulses of laser light are launched down the fiber, a small part of the transmitted light is backscattered by inhomogeneities in the fibre (a result of the manufacturing process). This backscattered light is then detected at the end of the fibre and after processing, a signal looking remarkably like seismic data is recorded.

A question often asked is, “what is this measurement?” with the answer expected to involve some sort of comparison to conventional sensors. In this paper we attempt to answer this question.

Fundamentals

A single measurement made using a DVS system is proportional to instantaneous strain at various positions down the fibre. As a seismic wave impinges on the fibre the strain on the fibre changes allowing the passage of the wave to be recorded (Figure 1).

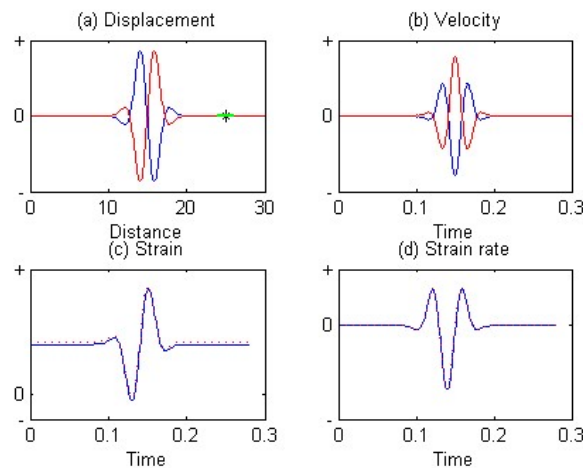


Figure 1 Simple example showing a *p*-wave travelling parallel to a section of the fibre (the green line) as (a) displacement, (b) velocity as measured by a geophone at the position of the black asterisk in (a), (c) strain measured by the fibre and (d) change in strain or strain rate. The red line is the result for a moving in the opposite direction (this line coincides with the blue line in (c) and (d)).

Unlike conventional measurements, strain is not measured at individual points but is instead measured across a section of the fibre; the length of the fibre being sensed is referred to as the *gauge length* or *differentiation interval*. Overlapping the intervals results in a channel spacing that is less than the interval itself (Figure 2).

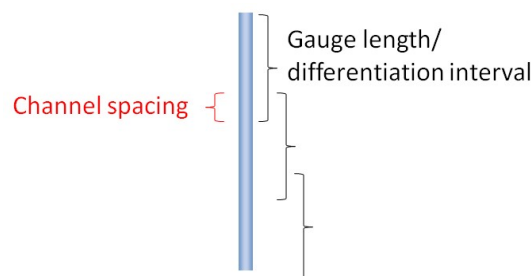


Figure 2 Diagram showing the gauge length or differentiation interval and the channel spacing.

Quantity

DVS systems measure either the variation in length of the fibre, which given a constant gauge length is equivalent to strain (the change in the length of the fibre divided by the length of the fibre at rest), or the strain rate which is the rate of change of strain with respect to time. The former is proportional to the sum of the displacements of all the segments of the fibre from their rest position while the latter is proportional to the average velocity of all the segments.

It is straightforward to convert 3-component recordings made using conventional sensors to a fibre response if they are dense enough. To do so you first convert the measurements to displacement and then calculate the length of the fibre as if the fibre had been attached to the sensors (Figure 3).

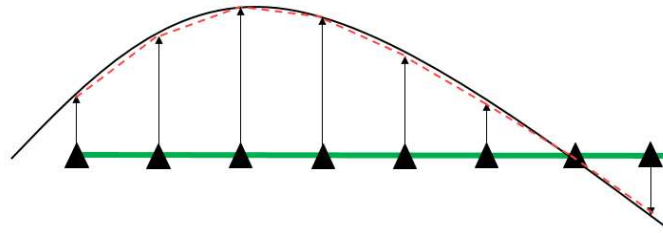


Figure 3 Diagrammatic representation of how to estimate the fibre response from geophone recordings. Each black triangle is a geophone, the black arrows are the displacement, the green line is the fibre length at rest and the sum of the red lines is the fibre length resulting from the stress of a passing S-wave.

Polarity

The strain gives no indication of the direction of the seismic wave that is causing it (see the red line in Figure 1). This results in the polarity of the wave being independent of the direction from which the wave is coming. Figure 4 (left) shows data recorded on the surface using a horizontal geophone and a section of fibre (see Papp et al. 2014 for further details on the acquisition), unlike the geophone traces, the polarity of the recorded signal is the same irrespective of the position of the source. Figure 4 (right) shows a section of three synthetic zero offset VSP records (geophone, fibre and hydrophone) containing the direction arrival and a reflection. The polarity of the first breaks are consistent but the polarity of the reflection for the fibre and hydrophone data is opposite to that of the geophone. Although the polarity of fibre measurements has not been formally defined, given its similarity to hydrophone measurements, it makes sense to adopt the SEG polarity convention for the latter as defined in Brown et al. (2002).

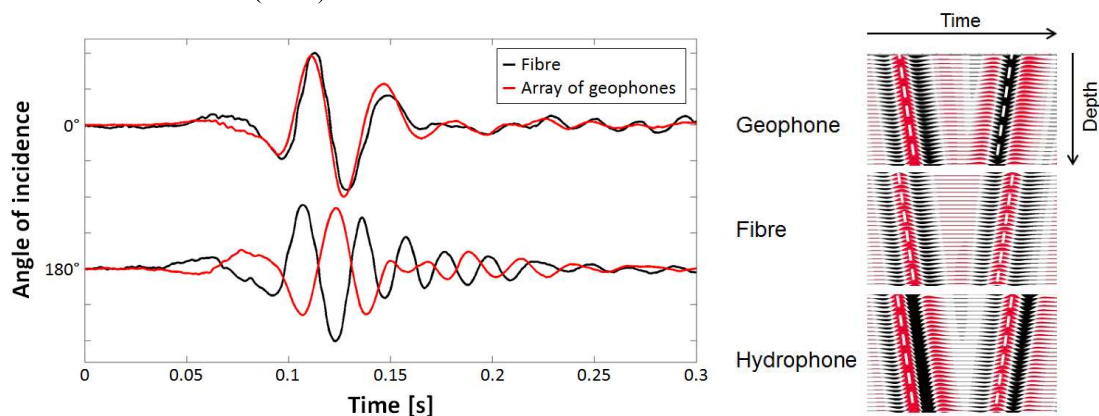
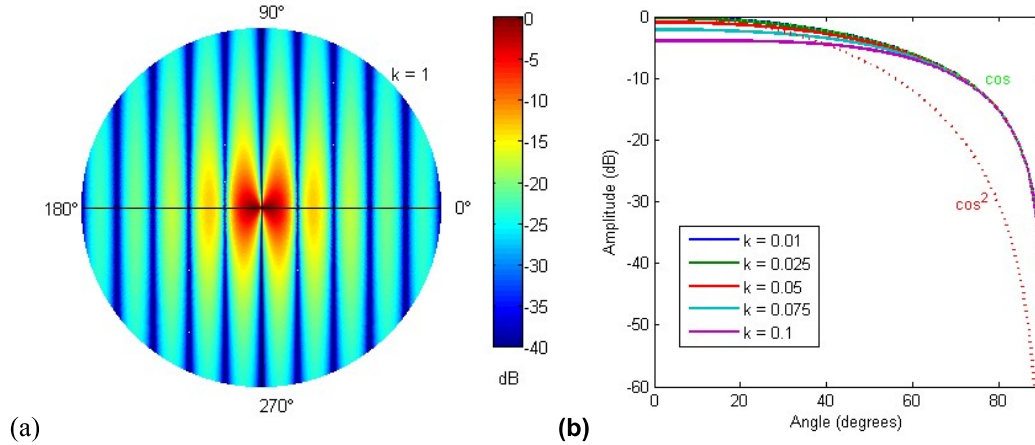


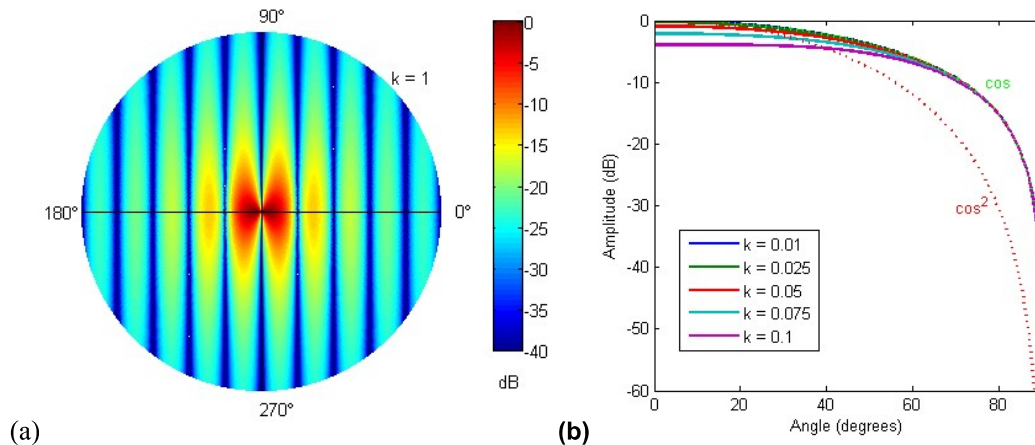
Figure 4 (Left) A comparison of surface recordings from a DVS and a horizontal geophone when the source is at opposite ends of the receiver line. (Right) Modelled zero-offset VSP data showing the polarity of the first arrival and a reflection (dashed lines).

Directionality

It is often stated that the magnitude of the amplitude measured using a DVS system is approximately proportional to $\cos^2\theta$ where θ is the angle of incidence, in contrast to a geophone where the measured amplitude is proportional to $\cos\theta$ (Figure 5), and a hydrophone which is omnidirectional. The relationship between incidence angle and amplitude is actually considerably more complex and is dependent on wavenumber as well (assuming that there is perfect coupling between the ground and the fibre).



(a) Figure 6a shows the amplitude vs. angle and wavenumber for a gauge length of 5 m; apart from the notch at $\theta = 90^\circ$ there are further notches where the apparent wavenumber is a multiple of the fibre-length.



(b) Figure 6b shows the amplitude versus angle for five different wavenumbers values ≤ 0.1 , as $k \rightarrow 0$ the relationship between azimuth and amplitude $\rightarrow \cos\theta$ but at other wavenumbers the amplitude is lower although the values all converge to the same values with increasing angle. For values of $k > 0.1$

the result is even more complex, for example at $k = 0.2$ (the dashed white line on

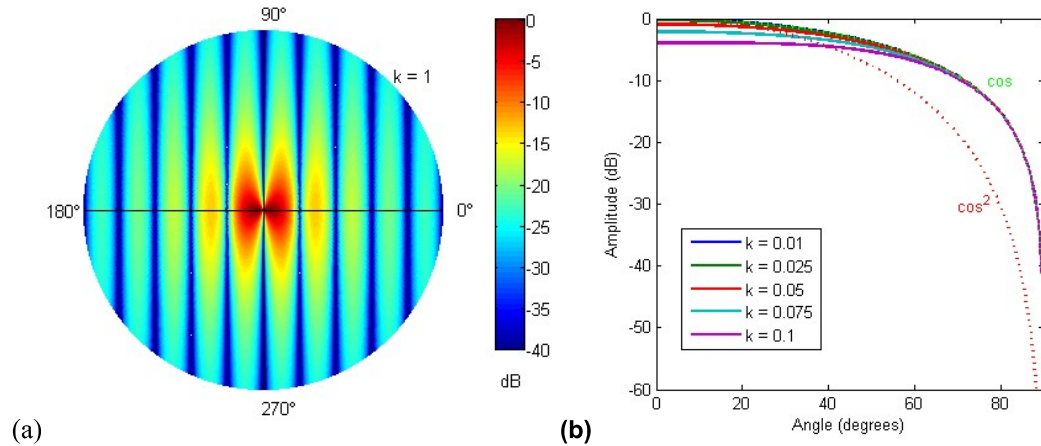


Figure 6a) the value is at a minimum at $\theta = 0^\circ$ before increasing to a maximum around 60° .

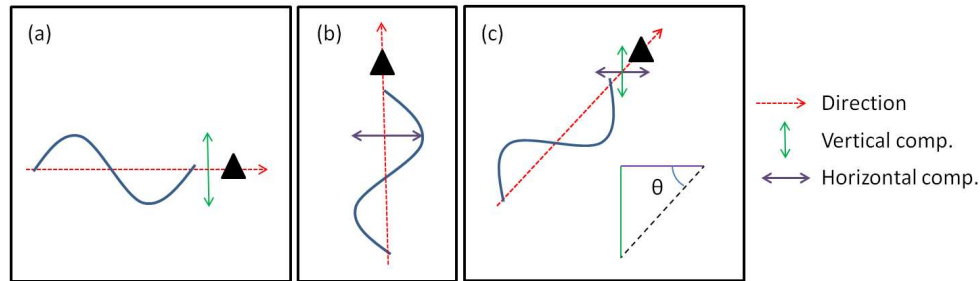


Figure 5 Simple schematic showing how the amplitude measured using a sensor (the black triangle) sensitive to movement in only a single direction (in this case vertically) is proportional to the cosine of the angle of incidence of the wave.

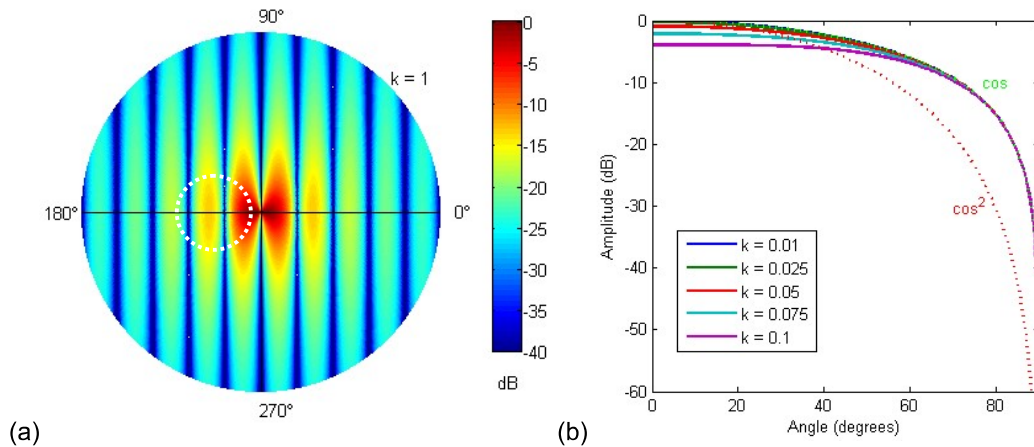


Figure 6 (a) A polar plot of the variation in amplitude with wavenumber and arrival angle for p-waves (all values are in dB) based on a fibre length of 5 m. (b) Gain vs. angle of incidence for five wavenumber values ≤ 0.1 , the $\cos$ and $\cos^2$ function are shown by dotted lines.

Figure 7 shows DVS VSP data recorded in a vertical well with a source offset of around 600 m. The red box highlights the area within which the direction arrival approach angle is approximately 90° and is therefore heavily attenuated.

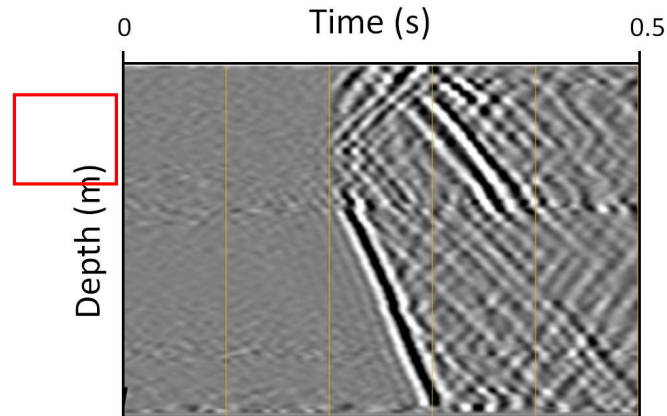


Figure 7 hDVS VSP data recorded with a source offset of around 600 m. The total depth is 670 m. Adapted from Frignet and Hartog (2014).

Discussion

In answer to the questions “what is this measurement?” our reply, as detailed above, is that a DVS system measures instantaneous strain, which is equivalent to the sum of the perturbations of each section of the gauge length from its mean position divided by its length. The time derivative of strain, the strain rate, is proportional to the average velocity measured at all points along the gauge length. DVS measurements have the polarity of a hydrophone measurement but the directionality is highly complex and depends on wavenumber as well as the angle of incidence.

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References

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